LITERATURE REVIEW: PREDICTIVE PLATFORM CONGESTION MODELLING FOR HIGH-DENSITY SUBURBAN RAIL NETWORKS

Synthesized Academic Literature Review matrix exported and structured from Elicit database | Core Research Publications (2017–2025)

Research Paper	Author(s)	Year	Objective	Methodology	Key Findings	Research Gap / Limitation
A Survey on Mumbai Suburban Local Train Travelers	Aditi Abhyankar, Aishwarya Narayanmorthy, V. Ramachandran, Mallika Mhapankar	**2012**	To examine the socio-economic role of the Mumbai suburban railway network and analyze commuter travel behavior across its three major corridors.	Field survey administered to regular daily commuters traveling distances between 10 to 60 km across Western, Central, and Harbour lines.	The suburban rail network serves as a critical economic lifeline for daily employment and education; high affordability (e.g., season passes) creates heavy, consistent commuter dependency requiring strategic transport management.	Focuses exclusively on socio-economic commuter demographics and general travel patterns, omitting platform physical capacity analysis, crowding dynamics, or predictive modeling.

LITERATURE REVIEW SUMMARY & METRICS

Metric Category	Value / Observation	Key Breakdown / Context
Total Studies Reviewed	**7 Papers**	Published between 2017 and 2025
Primary Target Variable	**Platform Crowding & Denied Boarding**	Short-term platform density (15-min ahead), denied boarding probability, and train/platform passenger loads.
Dominant Data Sources	**AFC (Smart Card) & AVL (Vehicle Tracking)**	Combined with GTFS-realtime, passenger arrival rates, and mobile location data (MD).
Primary Methodologies	**Supervised Learning, Bi-Level Regression & DRL**	Data-driven supervised learning, bi-level capacity-constrained loading, Deep Reinforcement Learning (DRL), and predictive optimization.
Case Study Locations	**Global Urban/Suburban Rail Networks**	New York City Subway (NYCT), Hong Kong MTR, Chicago CTA, Copenhagen Suburban Rail.
Core Operational Application	**Real-Time Decision Support & Control**	Holding trains, skipping stops, dynamic timetable rescheduling, passenger information displays, and demand management.